

Financial Frictions, Used Capital, and Misallocation

Job Market Paper

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Motivation

The Big Picture

- Low TFP is a key reason why developing countries are poorer
- Credit constraints prevent productive but poor entrepreneurs from starting or expanding
- Standard models assume firms use only new capital, so constrained firms can only operate at a smaller scale
- Micro data show that smaller and less wealthy firms often use **second-hand machines**

**Can used capital reduce misallocation and help
productive but poor people become entrepreneurs?**

- **Data:** Vietnamese manufacturing firms, 2005–2015
- **Model:** I add used capital to a standard model of entrepreneurial choice and capital allocation
- **Key trade-off:**
 - Used capital has a higher rental cost
 - But it uses less financing capacity because its market value is lower
- **Main findings:** access to used capital
 - Removes **39%** of the TFP loss caused by borrowing constraints
 - Closes **23%** of the entry shortfall for talented but poor entrepreneurs

**Used capital helps productive entrepreneurs
do more with the same wealth.**

Misallocation and financial frictions

Restuccia & Rogerson (2008); Hsieh & Klenow (2009); Buera, Kaboski & Shin (2011); Moll (2014)

Standard models study *how much* capital constrained firms use.

I also study *which type of capital they use: new or used.*

Entrepreneurship and wealth

Evans & Jovanovic (1989); Cagetti & De Nardi (2006); Buera & Shin (2013)

Used capital can change who enters entrepreneurship, not only firm scale.

Used capital and financing

Eisfeldt & Rampini (2007); Rampini (2019); Ma, Murfin & Pratt (2022)

Older assets require less financing because their market value is lower.

I bring this mechanism into a general-equilibrium model and quantify it using Vietnamese panel data.

Evidence on Used Capital

Sample

- Manufacturing firms, surveyed every two years: **5,131 firms** and **15,811 firm-year observations**

Key variables

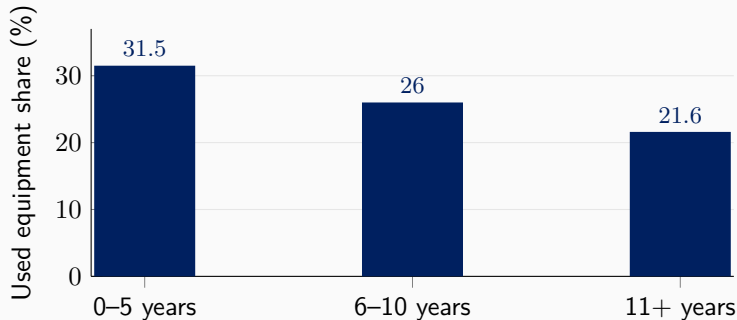
- Whether equipment is bought **new, used, or self-built**; its age; assets; debt; and credit access

Why Vietnam?

- A rare firm panel that records capital type, where used equipment is common (**24.5%** of equipment purchases) and credit constraints matter

**The data let me observe both firms' financial position
and their choice between new and used machines.**

Fact 1: Younger Firms Use More Used Equipment



Firms aged 0-5 use about 10 percentage points more used equipment than firms aged 11+.

Fact 2: Within-Firm Test

$$\text{UsedShare}_{it} = \beta_1 \text{ZeroDebt}_{it} + \beta_2 (\text{ZeroDebt}_{it} \times X_{it}) + \beta_3 X_{it} + \Gamma' Z_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

- **Outcome:** used equipment's share of the firm's main capital.
- X_{it} is either **log assets** or a **young-firm** indicator ($\text{age} \leq 5$).
- Controls: firm size and year effects; α_i compares each firm to itself over time.

Model prediction

For constrained firms, more collateral relaxes financing needs $\Rightarrow$ less reliance on used equipment.

Test: does a firm use less used equipment as its own assets grow?

Fact 2: Young and Low-Asset Firms Use More Used Capital

Within-firm estimates: each firm is compared with itself over time

Dep. var.: used-capital share (pp)	(1) Assets	(2) Young
Zero debt $\times$ log assets	-1.067** (0.462)	
Zero debt $\times$ young		3.885* (2.340)
Zero debt	-0.482 (0.836)	-1.046 (0.872)
Log assets	-0.260 (0.493)	-0.679 (0.453)
Young (age ≤ 5)		0.755 (1.658)
Observations	13,492	13,403
Firms	3,551	3,549
Firm and year FE	Yes	Yes
Size controls	Yes	Yes

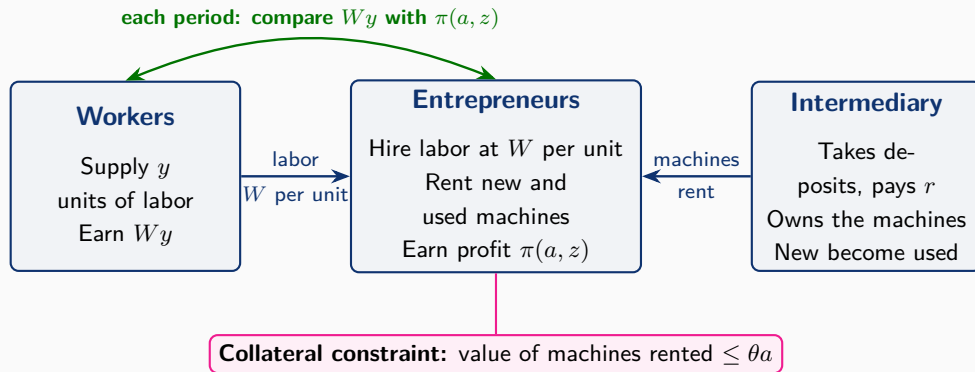
Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Young and low-asset, zero-debt firms use more used equipment.

Model

Who Is in the Model?

People differ in wealth a , entrepreneurial ability z , and worker ability y .



► Shocks

Preferences and Timing

- Time is discrete. People live forever and face individual shocks, but no aggregate shocks.
- Each period, a person chooses consumption today, c , and assets for next period, a' .
- They maximize lifetime utility:

$$U_0 = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \quad u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}.$$

- β measures patience and σ measures risk aversion.

Workers and entrepreneurs both choose consumption and saving.

- An entrepreneur with productivity z hires labor L and rents new and used capital
- Output is:

$$Y = \exp(z) (L^\alpha K^{1-\alpha})^\eta, \quad \eta < 1.$$

- K is made from new and used machines:

$$K = \left[\phi^{1/\gamma} K_n^{(\gamma-1)/\gamma} + (1-\phi)^{1/\gamma} K_u^{(\gamma-1)/\gamma} \right]^{\gamma/(\gamma-1)}.$$

- New and used capital are **imperfect substitutes**: γ determines how easily a firm substitutes between them, and ϕ their relative weight.

Firms choose the mix of capital, not only the amount.

Worker's Problem

- A worker with ability y supplies y efficiency units of labor.
- The wage per efficiency unit is W , so current labor income is:

$$I_w(y) = Wy.$$

- The worker chooses current consumption c and next period's assets a' :

$$c + a' = Wy + (1 + r)a, \quad a' \geq 0.$$

- Workers can save in deposits and earn the interest rate r , but they cannot borrow.
- People can switch between work and entrepreneurship every period

**A worker's value includes both today's wage income
and the option to become an entrepreneur later.**

Entrepreneur's Problem

- The entrepreneur maximizes operating profit:

$$\pi(a, z) = \max_{K_n, K_u, L \geq 0} \{Y - WL - p_n R_n K_n - p_u R_u K_u\}.$$

- **Financing constraint:**

$$\underbrace{p_n K_n}_{\text{new capital value}} + \underbrace{p_u K_u}_{\text{used capital value}} \leq \underbrace{\theta a}_{\text{financing capacity}}$$

- Wealth a is collateral: the machines an entrepreneur rents can be worth at most θa at market prices

Consumption, Saving, and Occupational Choice

Occupational choice

$$\Omega(a, z, y) = \max \left\{ V_e(a, z, y), V_w(a, z, y) \right\}$$

Same saving problem for both occupations $o \in \{e, w\}$:

$$V_o(a, z, y) = \max_{c, a' \geq 0} \left\{ u(c) + \beta E[\Omega(a', z', y') \mid z, y] \right\}$$

$$\text{s.t.} \quad c + a' = \underbrace{I_o}_{\text{only difference}} + (1 + r)a$$

Entrepreneur

$$I_e = \pi(a, z)$$

profits depend on wealth

Worker

$$I_w = Wy$$

wage $\times$ worker ability

► Details

Proposition 1 : Where Used Capital Comes From

- An intermediary owns all capital. It buys new machines and rents both new and used machines.
- Used machines come only from new machines that age:
 - a share ζ of new machines becomes used each period;
 - a share κ of used machines breaks and becomes worthless.

$$K'_n = (1 - \delta - \zeta)K_n + I, \quad K'_u = (1 - \delta - \kappa)K_u + \zeta K_n.$$

Proposition 1

The intermediary discounts at r , takes prices as given, and chooses investment to maximize the present value of rents net of investment. A stationary solution with $K_n, K_u > 0$ requires

$$R_n = r + \delta + \zeta \left(1 - \frac{p_u R_u}{r + \delta + \kappa} \right).$$

- Used capital's user cost: $R_u = r + \delta + \kappa$.

► Proof

Used machines rent at a higher rate ($R_u > R_n$) but are worth less ($p_u < p_n$).

Why Constrained Firms Use Used Capital

Every machine has two costs: the **rent** you pay, and the **collateral** it ties up.

	New machine	Used machine
Collateral it ties up (price)	1.00	0.65
Rent per year	0.12	0.15

- Used machines are **cheaper to buy**, but **more expensive to rent** (they break more).

Rich firm

- Collateral is not a problem
- Only rent matters $\Rightarrow$ mostly **new**

Poor firm

- Collateral runs out
- 100 of collateral: 100 new *or* 155 used machines
- $\Rightarrow$ tilts toward **used**, runs a bigger firm

As a firm gets richer, collateral matters less and it shifts back to new machines.

Used Capital and Financial Constraints

Goal: how does the financing constraint change the mix of new and used machines?

- Entrepreneur's problem with multiplier $\mu(a, z) \geq 0$ on the financing constraint:

$$\mathcal{L} = Y - WL - p_n R_n K_n - p_u R_u K_u + \mu(a, z) (\theta a - p_n K_n - p_u K_u)$$

- $\mu(a, z)$ = extra profit from \$1 more financing capacity (how tight collateral is).
- First-order conditions: each machine costs its rent plus the capacity it uses

$$MPK_n = p_n \left(\underbrace{R_n}_{\text{rent}} + \underbrace{\mu}_{\text{capacity}} \right), \quad MPK_u = p_u (R_u + \mu).$$

- Dividing them gives the mix $\rho \equiv K_u / K_n$:

$$\rho(\mu) = \frac{1 - \phi}{\phi} \left[\frac{p_n (R_n + \mu)}{p_u (R_u + \mu)} \right]^\gamma$$

- $\mu = 0$ (**rich firm**): compares *rents* $p_n R_n$ vs. $p_u R_u \Rightarrow$ same mix ρ^{uc} for all.
- μ large (**poor firm**): compares *collateral* p_n vs. $p_u \Rightarrow$ more used capital.

Used Capital and Financial Constraints: Collateral Productivity

Goal: what does a constrained firm gain from the mix of new and used machines?

- **Collateral productivity** Λ = capital per dollar of collateral:

$$\Lambda = \frac{K}{p_n K_n + p_u K_u}$$

- For a constrained firm, the financing constraint binds:

$$p_n K_n + p_u K_u = \theta a$$

- Multiply both sides by Λ :

$$K = \Lambda (p_n K_n + p_u K_u) \quad \Rightarrow \quad K = \underbrace{\Lambda}_{\text{mix}} \underbrace{\theta a}_{\text{wealth}}$$

- Wealth fixes θa : the firm cannot change it today.
- The choice of new vs. used decides Λ : the firm *can* change it.

Same wealth, better mix $\Rightarrow$ more capital.

Used Capital and Financial Constraints: Proposition 2

Goal: show that constrained firms tilt toward used capital, and what they gain from it.

Proposition 2 (Used-capital tilt and collateral relief)

If $R_u > R_n$ (equivalently $\kappa > \zeta(p_n - p_u)$), then the more constrained the firm (higher $\mu(a, z)$):

- it uses **more used capital** relative to new;
- it gets **more capital per dollar of collateral** (Λ rises), but the gain is bounded.

Intensive margin

constrained firms run at a larger scale

Extensive margin

more poor, productive people enter

► Details

Quantitative Results

Calibration: Vietnamese SMEs

Target moment	Data	Model
Debt-to-output	0.600	0.600
Top-decile employment share	0.632	0.633
Firm exit rate	0.098	0.097
Worker wage Gini	0.372	0.372
Real interest rate	0.031	0.031
New-to-used capital ratio	2.570	2.577
Used-share gradient with assets	-0.027	-0.027

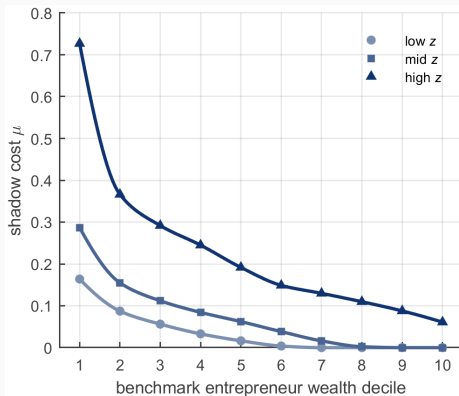
- **Comparison economy:** removes used capital and is recalibrated to match the same common moments.

Seven parameters, seven moments: the model matches all of them.

Mechanism in the Calibrated Economy

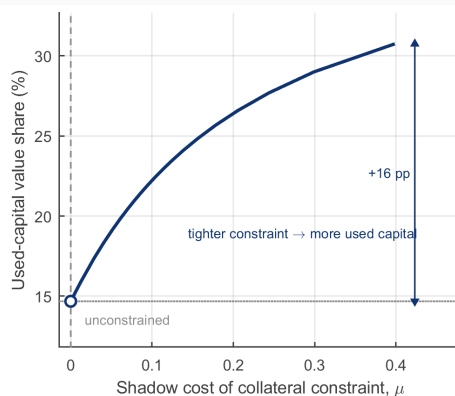
Who is constrained?

poor, productive firms have the highest μ



How do they respond?

higher $\mu \Rightarrow$ more used capital



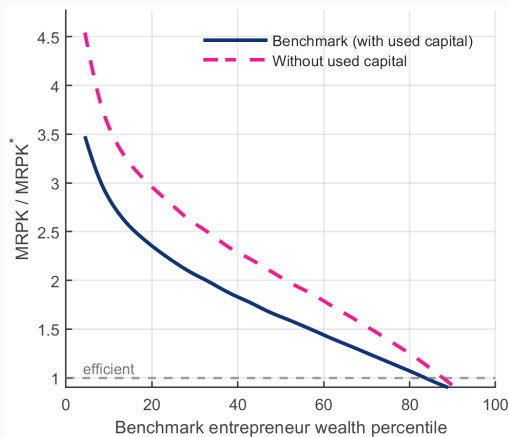
Result 1: Used Capital Reduces Misallocation

	Used capital	No used capital	Effect
TFP loss (log points)	5.72	9.39	39% lower
Average firm scale (K/K^*)	0.343	0.279	23% higher
Dispersion in MRPK	0.450	0.598	25% lower
Share of firms constrained	0.908	0.909	same

- Each economy is compared with its own frictionless version.
- Firms are **still constrained**: the share is the same in both economies.
- But each constrained firm gets **more capital from the same wealth**.

Used capital does not remove the constraint, but it makes it less costly:
39% of the TFP loss disappears.

Where the TFP Gain Comes From



- Poor entrepreneurs have the highest return to capital: they are the most constrained.
 - With used capital, their MRPK is lower: they operate with more capital.
 - The gain is largest at the bottom of the wealth distribution.
- ⇒ Capital moves toward its highest-return uses.

Used capital sends more capital to poor firms with the highest returns.

Result 2: Used Capital Changes Who Enters

	Used capital	No used capital	Difference
Talented, poor	42.0%	34.3%	+7.7 pp
Talented, rich	82.7%	83.3%	−0.6 pp

- Used machines let more **talented but poor** agents start a firm.
- Borrowing constraints keep fewer of them out: the entry gap falls by **23%**.
- Talented rich agents enter anyway: the constraint may limit their scale, but not their entry.

Who runs a firm depends less on wealth and more on talent.

► Details

Conclusion

- Constrained firms adjust **which** capital they use, not only how much.
- Used machines are costlier to rent but use **less of the financing limit**.
- In Vietnamese SMEs, firms rely less on used equipment as their own assets grow.
- Access to used capital removes **39%** of the TFP loss from borrowing constraints and expands entry by talented, poor entrepreneurs.

Second-hand equipment markets partly substitute for missing credit.

► Robustness

Thank You!

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Backup Slides

Evidence

- Empirics: additional checks

Model

- Idiosyncratic shocks
- Firm problem in closed form (1/2)
- Firm problem in closed form (2/2): μ
- Household problem

Intermediary (Proposition 1)

- The intermediary's problem
- Proof of Proposition 1
- Price vs. user cost; zero profit

Mechanism (Proposition 2)

- Capital mix and collateral productivity
- Where Proposition 2 comes from
- Proposition 2: proof sketch
- Three price comparisons

Equilibrium

- Aggregation and market clearing
- Stationary equilibrium

Calibration

- Calibration targets and fit
- Identifying the used-capital parameters
- Proposition 2 at benchmark prices

Results

- Loss measures and definitions
- Entry shortfall by wealth group

Robustness

- The result does not depend on one calibration
- Comparison designs: full results
- Robustness to ϕ and γ

Scope

- What the model abstracts from

Empirics: Additional Checks

Direct reports of binding constraints (formal loan applicants)

- Applicants reporting loan problems hold **11.7 pp** more used capital in 2013 (s.e. 3.10, $N = 644$); pooled 2011–2015, **6.8 pp** (s.e. 1.82, $N = 2,010$); similar with controls

Column (2): young $\times$ zero-debt interaction

- Raw reported age: 5.345 (2.378); firms observed in ≥ 3 rounds: 4.801 (2.439)
- Within-firm variation: ZeroDebt switches for 2,143 of 3,549 firms, Young for 1,021

Reading the evidence

- ZeroDebt marks self-financing, not rationing: most non-applicants did not need or want credit; 2.6% cite inadequate collateral
- No unconditional within-firm age gradient: Young = 0.755 (1.658) for firms with debt
- Attrition and the stock (not flow) nature of the vintage item work *against* finding an effect, and align the regressions with the model's value share

Idiosyncratic Shocks

Entrepreneurial productivity (persistent, with occasional redraws):

$$z' = \begin{cases} z, & \text{with probability } \omega, \\ \tilde{z}', & \tilde{z}' \sim F_z, \text{ with probability } 1 - \omega, \end{cases} \quad e^z \text{ has a Pareto right tail with parameter } \eta_p$$

Worker ability (AR(1) in logs, independent of z):

$$\log y' = \rho_y \log y + \sigma_y \varepsilon'_y, \quad \varepsilon'_y \sim N(0, 1), \quad \text{sd}(\log y) = \frac{\sigma_y}{\sqrt{1 - \rho_y^2}}$$

What disciplines each parameter

- $\omega \leftrightarrow$ firm exit rate 0.098 (switches from entrepreneurship to wage work)
- $\eta_p \leftrightarrow$ employment share of the top decile of firms, 0.632
- $\sigma_y \leftrightarrow$ worker wage Gini, 0.372; $\rho_y = 0.95$ assigned (Cagetti & De Nardi 2006)

Firm Problem in Closed Form (1/2): Given μ

Lagrangian of the firm's problem:

$$\mathcal{L} = e^z (L^\alpha K^{1-\alpha})^\eta - WL - p_n R_n K_n - p_u R_u K_u + \mu (\theta a - p_n K_n - p_u K_u)$$

For a given μ , an unconstrained problem with **effective unit prices**

$$c_j(\mu) = p_j (R_j + \mu), \quad j \in \{n, u\}$$

CES cost minimization: price of one unit of effective capital and conditional demands

$$P_K(\mu) = \left[\phi c_n(\mu)^{1-\gamma} + (1-\phi) c_u(\mu)^{1-\gamma} \right]^{\frac{1}{1-\gamma}}, \quad K_n = \phi \left(\frac{c_n}{P_K} \right)^{-\gamma} K, \quad K_u = (1-\phi) \left(\frac{c_u}{P_K} \right)^{-\gamma} K$$

Scale, labor, and output (decreasing returns, $\eta < 1$):

$$K^d(\mu) = \left[e^z \left(\frac{\alpha \eta}{W} \right)^{\alpha \eta} \right]^{\frac{1}{1-\eta}} \left[\frac{(1-\alpha) \eta}{P_K(\mu)} \right]^{\frac{1-\alpha \eta}{1-\eta}}, \quad L = \frac{\alpha \eta Y}{W}, \quad \frac{(1-\alpha) \eta Y}{K} = P_K(\mu)$$

Firm Problem in Closed Form (2/2): The Multiplier

Collateral used at μ is $K^d(\mu)/\Lambda(\rho(\mu))$. With $\Lambda^{uc} \equiv \Lambda(\rho(0))$, complementary slackness gives

$$\mu = 0 \text{ if } K^d(0) \leq \Lambda^{uc} \theta a; \quad \text{otherwise } \mu > 0 \text{ solves } K^d(\mu) = \Lambda(\rho(\mu)) \theta a$$

$K^d(\mu)$ falls with μ (P_K rises) and $\Lambda(\rho(\mu))$ rises (Prop. 2) $\Rightarrow$ a unique root, found by bisection.

Policies when the constraint binds:

$$K_n = \frac{\theta a}{p_n + p_u \rho(\mu)}, \quad K_u = \rho(\mu) K_n, \quad K = \Lambda(\rho(\mu)) \theta a, \quad L = \left[\frac{\alpha \eta e^z K^{(1-\alpha)\eta}}{W} \right]^{\frac{1}{1-\alpha\eta}}$$

$$\pi(a, z) = e^z \left(L^\alpha K^{1-\alpha} \right)^\eta - WL - p_n R_n K_n - p_u R_u K_u$$

- Marginal revenue product of effective capital = $P_K(\mu)$: **MRPK differs across firms only through μ**
- Two margins: scale (how far θa reaches) and mix ($\rho(\mu)$, how far each dollar stretches)

◀ Back

Household Problem and Occupational Choice

Preferences: $U_0 = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \quad u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}$

Values as an entrepreneur and as a worker:

$$V_e(a, z, y) = \max_{c \geq 0, a' \geq 0} \left\{ u(c) + \beta \mathbb{E}[\Omega(a', z', y') \mid z, y] \right\} \quad \text{s.t.} \quad c + a' = \pi(a, z) + (1+r)a$$

$$V_w(a, z, y) = \max_{c \geq 0, a' \geq 0} \left\{ u(c) + \beta \mathbb{E}[\Omega(a', z', y') \mid z, y] \right\} \quad \text{s.t.} \quad c + a' = Wy + (1+r)a$$

Occupational choice: $\Omega(a, z, y) = \max\{V_e(a, z, y), V_w(a, z, y)\}; \quad \text{entrepreneur} \iff V_e \geq V_w$

- $a' \geq 0$: no borrowing against labor income; the only credit is intermediated capital, limited by $p_n K_n + p_u K_u \leq \theta a$
- Occupation is a function of the current state, not a coordinate of the distribution $G(a, z, y)$
- No exogenous exit: the survey's exit rate maps to switches from running a firm to wage work

The Intermediary's Problem

The intermediary discounts at r , takes prices as given, and chooses investment:

$$V(K_n, K_u) = \max_{K'_n, I} \left\{ R_n K_n + p_u R_u K_u - I + \frac{V(K'_n, K'_u)}{1 + r} \right\}$$

$$\text{s.t.} \quad K'_n = (1 - \delta - \zeta) K_n + I, \quad K'_u = (1 - \delta - \kappa) K_u + \zeta K_n$$

Proposition 1 (capital prices)

A stationary solution with $K_n^s, K_u^s \in (0, \infty)$ requires

$$R_n = r + \delta + \zeta \left(1 - \frac{p_u R_u}{r + \delta + \kappa} \right)$$

- Only the product $p_u R_u$ enters this condition and any rental bill; the collateral constraint, written on market value, is what makes p_u and R_u matter separately
- Deposits are the only liability, capital the only asset: $D = p_n K_n^s + p_u K_u^s$

Proof of Proposition 1

Envelope conditions and the first-order condition for K'_n (primes denote next period):

$$V_n = R_n + (1 - \delta - \zeta) + \frac{\zeta}{1 + r} V'_u \quad (\text{A2})$$

$$V_u = p_u R_u + \frac{1 - \delta - \kappa}{1 + r} V'_u \quad (\text{A3})$$

$$0 = -1 + \frac{V'_n}{1 + r} \implies V'_n = 1 + r \quad (\text{A4})$$

In (A2), the continuation term $(1 - \delta - \zeta) V'_n / (1 + r)$ has already been replaced by $1 - \delta - \zeta$ using (A4).

Iterate (A3) forward and impose the steady state:

$$V_u = p_u R_u \sum_{s \geq 0} \left(\frac{1 - \delta - \kappa}{1 + r} \right)^s = p_u R_u \frac{1 + r}{r + \delta + \kappa} \quad (\text{A5})$$

Substitute (A4) and (A5) into (A2):

$$1 + r = R_n + 1 - \delta - \zeta + \zeta \frac{p_u R_u}{r + \delta + \kappa} \implies R_n = r + \delta + \zeta \left(1 - \frac{p_u R_u}{r + \delta + \kappa} \right) \quad \blacksquare$$

Splitting Price and User Cost; Zero Profit

No arbitrage on a used unit. Buy it at p_u , earn the rent $p_u R_u$, keep $(1 - \delta - \kappa) p_u$:

$$(1 + r) p_u = p_u R_u + (1 - \delta - \kappa) p_u \implies R_u = r + \delta + \kappa$$

The rent covers interest, depreciation, and breakage. Then $V_u = (1 + r) p_u$, and Proposition 1 becomes

$$R_n = r + \delta + \zeta (p_n - p_u) \iff p_u = p_n - \frac{R_n - r - \delta}{\zeta}$$

Zero profit. The technology is linear, so V is homogeneous of degree one. By Euler's theorem,

$$V = V_n K_n^s + V_u K_u^s = (1 + r) (p_n K_n^s + p_u K_u^s) = (1 + r) D$$

The intermediary is a veil: the economy behaves as if households held the capital stock at prices (p_n, p_u) .

Price adjustment. A higher R_n lowers p_u , which tilts demand toward used capital through both the rent and the collateral a unit absorbs.

Capital Mix and Collateral Productivity

Used-to-new ratio (the price ratio p_n/p_u does not depend on μ):

$$\rho(\mu) = \bar{\rho} \left[\frac{R_n + \mu}{R_u + \mu} \right]^\gamma, \quad \bar{\rho} \equiv \frac{1 - \phi}{\phi} \left(\frac{p_n}{p_u} \right)^\gamma, \quad \rho^{uc} \equiv \rho(0) = \bar{\rho} \left(\frac{R_n}{R_u} \right)^\gamma$$

Effective capital per dollar of collateral, using $K = K_n \Psi(\rho)$:

$$\Lambda(\rho) \equiv \frac{K}{p_n K_n + p_u K_u} = \frac{\Psi(\rho)}{p_n + p_u \rho}, \quad \Psi(\rho) \equiv \left[\phi^{1/\gamma} + (1 - \phi)^{1/\gamma} \rho^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}$$

$$\bar{\Lambda} = \Lambda(\bar{\rho}) = \left[\phi p_n^{1-\gamma} + (1 - \phi) p_u^{1-\gamma} \right]^{\frac{1}{\gamma-1}} > \frac{1}{p_n} > \Lambda(\rho=0) = \frac{\phi^{\frac{1}{\gamma-1}}}{p_n}$$

- $1/p_n$: the single-vintage economy (the recalibrated comparison)
- $\Lambda(\rho=0)$: a firm *denied* used capital (financing gain mixed with a missing-input penalty)
- Both types are used at any finite μ : $\partial K / \partial K_j \rightarrow \infty$ as $K_j \rightarrow 0$

Where Proposition 2 Comes From

Step 1: $\mu \uparrow \Rightarrow \rho \uparrow$ (from the mix)

$$\rho(\mu) = \frac{1-\phi}{\phi} \left(\frac{p_n}{p_u} \right)^\gamma \left[\frac{R_n + \mu}{R_u + \mu} \right]^\gamma, \quad \frac{d}{d\mu} \frac{R_n + \mu}{R_u + \mu} = \frac{R_u - R_n}{(R_u + \mu)^2} > 0 \text{ if } R_u > R_n.$$

ρ rises from $\rho^{uc} = \rho(0)$ to $\bar{\rho} = \frac{1-\phi}{\phi} (p_n/p_u)^\gamma$ as $\mu \rightarrow \infty$.

Step 2: $\rho \uparrow \Rightarrow \Lambda \uparrow$ (from collateral productivity)

$$\Lambda(\rho) = \frac{K}{p_n K_n + p_u K_u} = \frac{\Psi(\rho)}{p_n + p_u \rho}.$$

- $\bar{\rho}$ maximizes Λ : the most capital per \$1 of financing capacity.
- Λ rises for $\rho < \bar{\rho}$, and $\rho(\mu)$ moves from $\rho^{uc} < \bar{\rho}$ toward $\bar{\rho}$.
- Bounded: ρ never passes $\bar{\rho}$, so $\Lambda \leq \bar{\Lambda} = \Lambda(\bar{\rho})$.

Proposition 2: Proof Sketch

(i) **Tilt.** For given effective capital K the firm solves

$$\min_{K_n, K_u} p_n(R_n + \mu)K_n + p_u(R_u + \mu)K_u \quad \text{s.t.} \quad \left[\phi^{\frac{1}{\gamma}} K_n^{\frac{\gamma-1}{\gamma}} + (1-\phi)^{\frac{1}{\gamma}} K_u^{\frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}} = K$$

The FOCs give $\rho(\mu) = \bar{\rho} [(R_n + \mu)/(R_u + \mu)]^\gamma$, and

$$\frac{\partial}{\partial \mu} \frac{R_n + \mu}{R_u + \mu} = \frac{R_u - R_n}{(R_u + \mu)^2} > 0 \iff R_u > R_n, \quad \frac{R_n + \mu}{R_u + \mu} \xrightarrow{\mu \rightarrow \infty} 1 \Rightarrow \rho(\mu) \rightarrow \bar{\rho}$$

(ii) **Relief.** Maximizing $\Lambda(\rho) \iff$ minimizing collateral per unit of the bundle, $(p_n + p_u \rho)/\Psi(\rho)$:

- Strictly quasi-convex: $\rho \mapsto p_n + p_u \rho - t \Psi(\rho)$ is strictly convex for every $t > 0$ (Ψ strictly concave) $\Rightarrow$ unique minimizer $\bar{\rho}$ (user costs drop out as $\mu \rightarrow \infty$)
- $\rho(\mu)$ rises from $\rho^{uc} < \bar{\rho}$ toward $\bar{\rho}$, and Λ is increasing on $(0, \bar{\rho}) \Rightarrow \Lambda(\rho(\mu)) \uparrow \bar{\Lambda}$
- $\bar{\Lambda} > 1/p_n$ when $p_u < p_n$, $\gamma > 1$: then $p_u^{1-\gamma} > p_n^{1-\gamma}$, so the CES mean exceeds $p_n^{1-\gamma}$

Condition. $R_u > R_n \iff \kappa > \zeta(p_n - p_u)$; sufficient: $\zeta \leq \kappa \iff K_u^s/K_n^s \leq \kappa/(\delta + \kappa)$

Three Price Comparisons: Which One the Mechanism Uses

Comparison	Condition	What it delivers
Price per unit	$p_u < p_n \iff R_n > r + \delta$	$\bar{\Lambda} > 1/p_n$: more capital per collateral dollar
User cost per \$ of value	$R_u > R_n \iff \kappa > \zeta(p_n - p_u)$	the tilt, $\rho'(\mu) > 0$
Rent per unit	$p_u R_u \gtrless p_n R_n$	nothing: no result depends on it

Read the effective price $p_j(R_j + \mu)$ per rental dollar. A dollar of rent on type j ties up

$$\frac{p_j}{p_j R_j} = \frac{1}{R_j}$$

of financing capacity. The price cancels, so on this margin **the user cost alone decides**: renting used capital frees collateral relative to new exactly when $R_u > R_n$.

At the calibration: $p_u = 0.645 < 1$, $R_u = 0.234 > R_n = 0.119$, and $p_u R_u \approx 0.151 > p_n R_n = 0.119$.

Both $p_u < p_n$ and $R_u > R_n$ are equilibrium properties, not restrictions on primitives.

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Aggregation and Market Clearing

Entrepreneurs and workers: $\mathcal{E} = \{(a, z, y) : V_e \geq V_w\}$, $\mathcal{W} = \{(a, z, y) : V_e < V_w\}$

$$K_n^d = \int_{\mathcal{E}} K_n(a, z) dG, \quad K_u^d = \int_{\mathcal{E}} K_u(a, z) dG, \quad L^d = \int_{\mathcal{E}} L(a, z) dG, \quad L^s = \int_{\mathcal{W}} y dG$$

Supply: stationary used stock and the balance sheet, with $D = \int a dG$:

$$\zeta K_n^s = (\delta + \kappa) K_u^s, \quad D = p_n K_n^s + p_u K_u^s \implies K_n^s = \frac{D}{p_n + p_u \zeta / (\delta + \kappa)}, \quad K_u^s = \frac{\zeta}{\delta + \kappa} K_n^s$$

Market clearing: $K_n^d = K_n^s$, $K_u^d = K_u^s$, $L^d = L^s$. Three prices, three markets:

- W clears labor; r clears new capital through deposits; R_n adjusts until $K_n^d/K_u^d = (\delta + \kappa)/\zeta$, after which used-capital clearing is automatic

External finance (the moment that disciplines θ):

$$B^{ext} = \int_{\mathcal{E}} \max\{0, p_n K_n(a, z) + p_u K_u(a, z) - a\} dG, \quad B^{ext}/Y = 0.600$$

Stationary Competitive Equilibrium

Prices (W, r, R_n) with implied $p_u = p_n - (R_n - r - \delta)/\zeta$ and $R_u = r + \delta + \kappa$; value functions $\{V_e, V_w, \Omega\}$; policies; firm input demands; stocks (K_n^s, K_u^s) ; and a distribution $G(a, z, y)$ such that:

1. given (W, R_n, R_u, p_u) , firm input demands solve the profit problem subject to the collateral constraint;
2. given prices and Ω , consumption, saving, and occupation solve the household problem;
3. the intermediary's investment is optimal given (r, R_n) : Proposition 1 holds, profit is zero, and $D = p_n K_n^s + p_u K_u^s$;
4. G is invariant under the optimal policies and the processes for z and y ;
5. the labor, new-capital, and used-capital markets clear.

Benchmark	W	r	R_n	p_u	R_u	Entrepreneurs	Constrained
Equilibrium	0.540	0.031	0.119	0.645	0.234	0.221	0.908

Calibration Targets and Fit

Moment	Param.	Data	Model	Source
Debt-to-output, B^{ext}/Y	θ	0.600	0.600	Debt over value added, 2013 & 2015 SME rounds
Top-decile employment share	η_p	0.632	0.633	2013 SME round, paid employment
Annual firm exit rate	ω	0.098	0.097	SME survey; cf. 9–10% in Berkel et al. (2020)
Worker wage Gini	σ_y	0.372	0.372	VHLSS 2010 (Doan et al. 2023)
Real interest rate	β	0.031	0.031	Deposit rate less GDP-deflator inflation, 2013 (WDI)
New-to-used capital ratio	ζ	2.570	2.577	SME survey
Used-share gradient in log assets	κ	−0.027	−0.027	SME survey, 2013 cross-section

- Model column: benchmark. The no-used-capital economy matches the five common moments
- $\beta = 0.858$ is what clears the deposit market at $r = 3.1\%$ once entrepreneurs earn a return premium; close to the 0.865 in Cagetti & De Nardi (2006)
- Exit: the survey records closure or suspension of the household's enterprise, which is what a switch to wage work is in the model

Identifying the Used-Capital Parameters

Relative quantity (stationary stocks):

$$\frac{K_n^s}{K_u^s} = \frac{\delta + \kappa}{\zeta} = 2.570$$

Relative collateral price

(firms with a positive used share, 2013):

$$s_{u,i} = b_0 + b_1 \log A_i + \varepsilon_i$$

$$b_1 = -0.027 \text{ (0.0061)}, \quad N = 886$$

Jointly: $\kappa = 0.143$, $\zeta = 0.079$

$\Rightarrow R_u = 0.234 > R_n = 0.119$, and $\zeta \leq \kappa$

holds

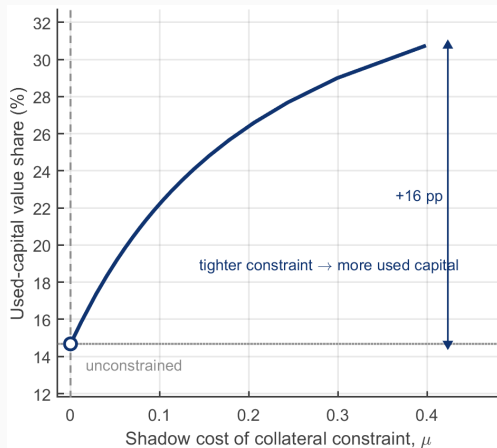
Why the gradient pins κ

- A bigger collateral advantage of used capital $\Rightarrow$ small, constrained firms lean on it more $\Rightarrow$ a steeper (more negative) b_1
- A between-firm size gradient in one cross-section, natural for a stationary model; distinct from the within-firm interactions in Table 2

Assigned CES parameters

- $\phi = 0.70$ is weakly identified: ζ delivers the aggregate ratio at any ϕ . At $\phi = 0.75$ (Lanteri 2018) the results are unchanged
- $\gamma = 2$: Edgerton (2011) estimates 1.7–2.0 for farm and 1.9–2.4 for construction machinery

Proposition 2 at Benchmark Prices



At $p_u = 0.645$, $R_n = 0.119$, $R_u = 0.234$, $\phi = 0.7$, $\gamma = 2$:

μ	K_u/K_n	Used value share	$\Lambda(\rho(\mu))$
0	0.267	14.7%	1.069
0.1	0.443	22.2%	1.122
0.2	0.557	26.4%	1.141
0.4	0.691	30.8%	1.155
∞	1.030	39.9%	1.165

- Steep at small μ : nearly half the rise to $\mu = 0.4$ is done by $\mu = 0.1$
- Bounded relief: Λ at most 9% above the unconstrained mix
- Same $\mu \Rightarrow$ same mix: a financing response, not technology

Loss Measures and Definitions

Frictionless counterpart. In each economy, raise θ until the constraint is slack for every operating firm, hold all other parameters fixed, and recompute prices, occupations, and the stationary distribution. MRPK dispersion is then numerically zero.

$$\mathcal{M} = 100 \log \frac{\text{TFP}^{\text{frictionless}}}{\text{TFP}^{\text{friction}}}, \quad \text{share removed} = \frac{\mathcal{M}^{\text{no used}} - \mathcal{M}^{\text{bm}}}{\mathcal{M}^{\text{no used}}} = \frac{9.39 - 5.72}{9.39} = 39.1\%$$
$$S = 100 \log \frac{e^{\text{frictionless}}}{e^{\text{friction}}}, \quad \text{share repaired} = 1 - \frac{S^{\text{bm}}}{S^{\text{no used}}} = 1 - \frac{53.2}{68.7} = 22.6\%$$

- e : entrepreneurship rate of poor talented agents (top decile of z , wealth below the benchmark median; cutoffs built once in the benchmark and applied in levels to both economies)
- K/K^* : a firm's new capital relative to what it would install if the constraint did not bind, at the economy's prices; averaged across firms, very high ratios capped
- MRPK^* (Figure 2): the economy-wide average MRPK in the frictionless economy

Details: Entry Shortfall by Wealth Group

Entry shortfall: $S = 100 \log \left(\frac{e^{\text{frictionless}}}{e^{\text{friction}}} \right)$ (how much the constraint lowers entry)

Talented = top-decile z	Used capital	No used capital
Poor (wealth below the median)		
Entry rate, with the constraint	0.420	0.343
Entry rate, frictionless	0.714	0.682
Shortfall S (log points)	53.2	68.7
Rich (wealth above the median)		
Entry rate, with the constraint	0.827	0.833
Entry rate, frictionless	0.801	0.744
Shortfall S (log points)	-3.2	-11.3

- **Poor:** used capital shrinks the entry gap by $1 - \frac{53.2}{68.7} = 23\%$.
- **Rich:** negative shortfall, a general-equilibrium wage effect. Removing the constraint raises labor demand and the wage, so some marginal high- z agents switch to wage work.

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Robustness: The Result Does Not Depend on One Calibration

Comparison economy without used capital	TFP loss	Gain from used capital
Productivity distribution held fixed	7.11	19.5%
Fully recalibrated (headline)	9.39	39.1%

- The quantitative size depends on the comparison: used capital removes **20–39%** of the TFP loss.
- It repairs **23–32%** of the lost entry by talented, poor agents.
- Across all specifications, the conclusion is unchanged:

Used capital raises scale, lowers misallocation, and expands entry by talented poor agents.

► All designs

► ϕ, γ

Comparison Designs: Full Results

Economy	η_p	$\mathcal{M}$ (log pts)	Misalloc. reduced	Entry repaired	K/K^*	sd(log MRPK)
Benchmark (used capital)	5.949	5.72	—	—	0.343	0.450
No used capital, tail pinned	5.949	7.11	19.5%	32.0%	0.268	0.530
Tail diagnostic	5.949	7.42	22.9%	27.8%	0.295	0.530
No used capital, recalibrated	5.397	9.39	39.1%	22.6%	0.279	0.598

- **Tail pinned:** $(\eta_p, \omega, \sigma_y)$ at benchmark values; only (θ, β) recalibrated
- **Tail diagnostic:** recalibrated economy with η_p back at 5.949; prices re-cleared, no moment refit
- Restoring η_p alone moves $\mathcal{M}$ from 9.39 to 7.42; $(\theta, \omega, \sigma_y, \beta)$ and prices account for the other 0.31
- The margins move in opposite directions as the tail is refit: a fatter tail adds very productive entrepreneurs, costly to constrain (misallocation share rises) but inframarginal for entry (entry share falls)

Misalloc. reduced = $1 - \mathcal{M}^{\text{bm}}/\mathcal{M}$; entry repaired = $1 - \mathcal{S}^{\text{bm}}/\mathcal{S}$; each economy against its own frictionless twin.

Robustness to the Assigned CES Parameters

	Benchmark ($\phi = 0.70, \gamma = 2$)	$\phi = 0.75$	$\gamma = 1.7$	$\gamma = 2.5$
<i>Gain from removing the collateral constraint</i>				
TFP loss $\mathcal{M}$ (log points)	5.72	5.74	5.75	5.70
Misallocation reduced (%)	39.1	38.9	38.7	39.3
<i>Allocation in the constrained equilibrium</i>				
Mean scale, K/K^*	0.343	0.350	0.347	0.337
Share of firms constrained	0.908	0.906	0.907	0.908
Entrepreneur share	0.221	0.220	0.221	0.220
<i>Cleared prices</i>				
Wage W	0.540	0.542	0.540	0.540
Interest rate r	0.031	0.031	0.031	0.031
New-capital rental R_n	0.119	0.124	0.119	0.120
Used-capital price p_u	0.645	0.582	0.648	0.640

Each column re-clears (W, r, R_n) holding $(\theta, \eta_p, \omega, \kappa, \zeta, \sigma_y, \beta)$ at benchmark values: sensitivity sweeps, not recalibrations. $\mathcal{M}$ stays between 5.70 and 5.75 log points.

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What the Model Abstracts From

Deliberately outside the model

- Supply of used capital is reduced form: new capital ages at rate ζ , used capital breaks at rate κ , and the intermediary prices both at zero profit
- No resale decision by firms, and no illiquidity or adverse selection in the used market (cf. Lanteri 2018; Lanteri & Rampini 2023; Bierdel, Drenik, Herreño & Ottonello 2023)
- No timing-of-payments channel (Eisfeldt & Rampini 2007); firms rent capital services
- Stationary economy: no aggregate uncertainty, no transition dynamics

What the mechanism needs

- Collateral counted at **market value**: $p_n K_n + p_u K_u \leq \theta a$
- Used capital with a higher user cost per dollar of value: $R_u > R_n$

⇒ Under these two, Proposition 2 delivers the tilt and the collateral relief, whatever produces p_u and R_u